Lines, Triangles & Polygons

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational] • Target: Similar triangles and proportional reasoning**

Triangle ABC shown below has $AB = 12$, $BC = 16$, and $CA = 20$. It is similar to triangle DEF, where $DE = 15$ corresponds to AB. Find EF and FD.

Scale factor $= \frac{DE}{AB} = \frac{15}{12} = 1.25$. $EF = BC \times 1.25 = 16 \times 1.25 = 20$. $FD = CA \times 1.25 = 20 \times 1.25 = 25$.

2 **[Foundational] • Target: The Law of Cosines**

In triangle PQR shown below, $PQ = 9$, $QR = 13$, and $\angle Q = 50^\circ$. Find $PR = x$, to the nearest hundredth.

$x^2 = 9^2 + 13^2 - 2(9)(13)\cos 50^\circ \approx 250 - 150.41 = 99.59$, so $x \approx \sqrt{99.59} \approx 9.98$.

3 **[Foundational] • Target: Area of polygons**

The trapezoid below has parallel sides of length 14 and 8, and height 6. Find its area.

Area $= \frac{1}{2}(14 + 8)(6) = \frac{1}{2}(22)(6) = 66$ square units.

4 **[Foundational] • Target: Parallel lines and transversals**

Two parallel lines are cut by a transversal, forming an angle of 105° on one line. Find the corresponding angle, the alternate interior angle, and the co-interior (same-side interior) angle on the other line.

Corresponding angle $= 105^\circ$ (equal). Alternate interior angle $= 105^\circ$ (equal). Co-interior angle $= 180^\circ - 105^\circ = 75^\circ$ (supplementary).

5 **[Foundational] • Target: Parallel lines and transversals**

Two angles formed by a transversal crossing parallel lines are alternate exterior angles, so they are equal. If one measures $3x + 10$ and the other measures $5x - 30$, find x and the measure of the angle.

$3x + 10 = 5x - 30 \Rightarrow 40 = 2x \Rightarrow x = 20$. The angle measures $3(20) + 10 = 70^\circ$.

6 **[Foundational] • Target: Parallel lines and transversals**

Two parallel lines are cut by a transversal. One interior angle is $4x + 15$, and its corresponding angle on the other line is $6x - 25$. Find x and the angle measure.

Corresponding angles are equal: $4x + 15 = 6x - 25 \Rightarrow 2x = 40 \Rightarrow x = 20$. The angle measures $4(20) + 15 = 95^\circ$.

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7 [Foundational] • Target: Triangle angle sum and the exterior angle theorem

The angles of a triangle are in the ratio $3:5:7$. Find each angle.

The parts sum to $3+5+7=15$, and the total is 180°, so each part is 12°. The angles are 36°, 60°, and 84°.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Triangle angle sum and the exterior angle theorem

In triangle ABC, $\angle A = 52^\circ$ and the exterior angle at C is 130°. Find $\angle B$.

The exterior angle equals the sum of the remote interior angles: $130^\circ = 52^\circ + \angle B \Rightarrow \angle B = 78^\circ$.

9 [Intermediate] • Target: Triangle angle sum and the exterior angle theorem

An exterior angle of a triangle measures 115°, and one remote interior angle is 50°. Find the other remote interior angle and the third (adjacent) interior angle.

The other remote interior angle is $115^\circ - 50^\circ = 65^\circ$. The adjacent interior angle is supplementary to the exterior angle: $180^\circ - 115^\circ = 65^\circ$. Check: $50^\circ + 65^\circ + 65^\circ = 180^\circ$.

10 [Intermediate] • Target: Triangle congruence criteria

State which congruence criterion (SSS, SAS, ASA, AAS, or RHS) proves triangles PQR and XYZ congruent, given $PQ = XY$, $\angle Q = \angle Y$, and $QR = YZ$.

SAS — two sides and the included angle.

11 [Intermediate] • Target: Triangle congruence criteria

Triangle DEF has $DE = 9$, $EF = 12$, $\angle E = 90^\circ$, and triangle GHI has $GH = 9$, $HI = 12$, $\angle H = 90^\circ$. Are the triangles congruent? By which criterion?

Yes — congruent by SAS, since two sides and the included right angle are equal in both triangles.

12 [Intermediate] • Target: Triangle congruence criteria

Two triangles have all three pairs of corresponding sides equal in length. Which congruence criterion applies, and are the triangles necessarily congruent?

SSS applies, and yes, the triangles are necessarily congruent.

13 [Intermediate] • Target: Similar triangles and proportional reasoning

Triangle $ABC \sim$ triangle DEF, with $AB = 6$, $BC = 8$, and $DE = 9$. Find EF.

Scale factor $= \frac{DE}{AB} = \frac{9}{6} = 1.5$. $EF = 8 \times 1.5 = 12$.

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14 [Intermediate] • Target: Similar triangles and proportional reasoning

Triangle JKL has $JK = 15$ and $KL = 21$, and is similar to triangle MNO with scale factor $\frac{2}{3}$ (triangle MNO is smaller), where MN corresponds to JK. Given $MN = 10$, verify the scale factor is consistent, then find NO.

Check: $\frac{MN}{JK} = \frac{10}{15} = \frac{2}{3}$, consistent. $NO = KL \times \frac{2}{3} = 21 \times \frac{2}{3} = 14$.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: The Law of Cosines • Cross-Lesson: 5.01 Trigonometry

A triangle has sides $a = 7$, $b = 8$, $c = 10$. Use the Law of Cosines to find angle C (opposite side c), to the nearest tenth of a degree.

$\cos C = \frac{a^2 + b^2 - c^2}{2ab} = \frac{49 + 64 - 100}{112} = \frac{13}{112} \approx 0.1161$. $C \approx 83.3^\circ$.

16 [Challenging] • Target: The Law of Cosines • Cross-Lesson: 6.01 Circles & Coordinate Geometry

A triangle has sides $a = 6$, $b = 6$, and included angle $C = 80^\circ$. Find side c, to the nearest hundredth, and state what type of triangle this is based on the two given equal sides.

$c^2 = 6^2 + 6^2 - 2(6)(6)\cos 80^\circ = 72 - 72(0.1736) \approx 59.50$, so $c \approx 7.71$. Since two sides are equal, the triangle is isosceles.

17 [Challenging] • Target: Heron's formula • Cross-Lesson: 7.01 Solid Geometry & Similarity

Find the area of a triangle with sides 13, 14, 15 using Heron's formula.

$s = \frac{13 + 14 + 15}{2} = 21$. Area $= \sqrt{21(21 - 13)(21 - 14)(21 - 15)} = \sqrt{21 \cdot 8 \cdot 7 \cdot 6} = \sqrt{7056} = 84$.

18 [Challenging] • Target: Heron's formula • Cross-Lesson: 8.01 Statistics & Probability

A triangular plot of land has sides 9, 10, 17. Find its area using Heron's formula.

$s = \frac{9 + 10 + 17}{2} = 18$. Area $= \sqrt{18(18 - 9)(18 - 10)(18 - 17)} = \sqrt{18 \cdot 9 \cdot 8 \cdot 1} = \sqrt{1296} = 36$.

19 [Challenging] • Target: Heron's formula • Cross-Lesson: 1.01 Advanced Algebra

A triangular sail has sides 5 m, 12 m, and 13 m. Find its area using Heron's formula, and verify using the right-triangle area formula (since 5-12-13 is a Pythagorean triple).

$s = \frac{5 + 12 + 13}{2} = 15$. Area $= \sqrt{15 \cdot 10 \cdot 3 \cdot 2} = \sqrt{900} = 30$ m^2. Check: since it's a right triangle with legs 5 and 12, Area $= \frac{1}{2}(5)(12) = 30$ m^2. Matches.

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20 [Challenging] • Target: Interior and exterior angles of polygons • Cross-Lesson: 2.01 Functions: Composite, Inverse & Logarithmic

Find the sum of the interior angles of a regular decagon (10 sides), and the measure of each interior angle.

Sum $= (10 - 2) \times 180^\circ = 1440^\circ$. Each interior angle $= \frac{1440^\circ}{10} = 144^\circ$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Parallel lines and transversals

The angles of a triangle are in the ratio $2 : 1 : 3$. Find each angle.

a. $30, 64, 86$

b. $30, 60, 90$ ✓

c. $32, 60, 88$

d. $90, 60, 30$

The parts sum to $2 + 1 + 3 = 6$, and the total is 180°, so each part is 30°. The angles are 30°, 60°, and 90°.

22 [Foundational] • Target: Triangle angle sum and the exterior angle theorem

The angles of a triangle are in the ratio $9 : 3 : 3$. Find each angle.

a. $38, 36, 106$

b. $36, 40, 104$

c. $36, 36, 108$ ✓

d. $108, 36, 36$

The parts sum to $9 + 3 + 3 = 15$, and the total is 180°, so each part is 12°. The angles are 36°, 36°, and 108°.

23 [Foundational] • Target: Triangle congruence criteria

The angles of a triangle are in the ratio $1 : 5 : 6$. Find each angle.

a. $90, 75, 15$

b. $15, 79, 86$

c. $15, 75, 90$ ✓

d. $17, 75, 88$

The parts sum to $1 + 5 + 6 = 12$, and the total is 180°, so each part is 15°. The angles are 15°, 75°, and 90°.

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24 [Foundational] • Target: Similar triangles and proportional reasoning

The angles of a triangle are in the ratio $4:5:6$. Find each angle.

a. $50, 60, 70$

b. $48, 64, 68$

c. $48, 60, 72$ ✓

d. $72, 60, 48$

The parts sum to $4+5+6=15$, and the total is 180°, so each part is 12°. The angles are 48°, 60°, and 72°.

25 [Foundational] • Target: The Law of Cosines

The angles of a triangle are in the ratio $3:3:9$. Find each angle.

a. $108, 36, 36$

b. $38, 36, 106$

c. $36, 40, 104$

d. $36, 36, 108$ ✓

The parts sum to $3+3+9=15$, and the total is 180°, so each part is 12°. The angles are 36°, 36°, and 108°.

26 [Foundational] • Target: Heron's formula

The angles of a triangle are in the ratio $9:5:6$. Find each angle.

a. $45, 54, 81$ ✓

b. $81, 54, 45$

c. $47, 54, 79$

d. $45, 58, 77$

The parts sum to $9+5+6=20$, and the total is 180°, so each part is 9°. The angles are 45°, 54°, and 81°.

27 [Foundational] • Target: Interior and exterior angles of polygons

The angles of a triangle are in the ratio $5:6:1$. Find each angle.

a. $15, 75, 90$ ✓

b. $17, 75, 88$

c. $90, 75, 15$

d. $15, 79, 86$

The parts sum to $5+6+1=12$, and the total is 180°, so each part is 15°. The angles are 15°, 75°, and 90°.

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28 [Foundational] • Target: Area of polygons

An exterior angle of a triangle measures 82°, and one remote interior angle is 30°. Find the other remote interior angle.

a. 52° ✓

b. 57°

c. 47°

d. 30°

The exterior angle equals the sum of the remote interior angles: $82^\circ = 30^\circ + \angle B \Rightarrow \angle B = 52^\circ$.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: A well-designed word problem: a surveyor's triangular lots

An exterior angle of a triangle measures 76°, and one remote interior angle is 44°. Find the other remote interior angle.

a. 27°

b. 44°

c. 37°

d. 32° ✓

The exterior angle equals the sum of the remote interior angles: $76^\circ = 44^\circ + \angle B \Rightarrow \angle B = 32^\circ$.

30 [Intermediate] • Target: Coordinate Geometry of Lines

An exterior angle of a triangle measures 56°, and one remote interior angle is 28°. Find the other remote interior angle.

a. 28° ✓

b. 31°

c. 23°

d. 33°

The exterior angle equals the sum of the remote interior angles: $56^\circ = 28^\circ + \angle B \Rightarrow \angle B = 28^\circ$.

31 [Intermediate] • Target: Triangle Theorems & Inequalities

An exterior angle of a triangle measures 54°, and one remote interior angle is 23°. Find the other remote interior angle.

a. 23°

b. 36°

c. 31° ✓

d. 26°

The exterior angle equals the sum of the remote interior angles: $54^\circ = 23^\circ + \angle B \Rightarrow \angle B = 31^\circ$.

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32 [Intermediate] • Target: Polygons & Interior/Exterior Angles

An exterior angle of a triangle measures 88°, and one remote interior angle is 22°. Find the other remote interior angle.

a. 22°

b. 71°

c. 61°

d. 66° ✓

The exterior angle equals the sum of the remote interior angles: $88^\circ = 22^\circ + \angle B \Rightarrow \angle B = 66^\circ$.

33 [Intermediate] • Target: Geometric Transformations & Dilations

An exterior angle of a triangle measures 86°, and one remote interior angle is 54°. Find the other remote interior angle.

a. 32° ✓

b. 54°

c. 27°

d. 37°

The exterior angle equals the sum of the remote interior angles: $86^\circ = 54^\circ + \angle B \Rightarrow \angle B = 32^\circ$.

34 [Intermediate] • Target: Parallel lines and transversals

An exterior angle of a triangle measures 47°, and one remote interior angle is 25°. Find the other remote interior angle.

a. 25°

b. 22° ✓

c. 27°

d. 17°

The exterior angle equals the sum of the remote interior angles: $47^\circ = 25^\circ + \angle B \Rightarrow \angle B = 22^\circ$.

35 [Intermediate] • Target: Triangle angle sum and the exterior angle theorem

A triangle has sides $a = 12$, $b = 10$, and included angle $C = 40^\circ$. Find side c, to the nearest hundredth.

a. 8.26

b. 8.76

c. 7.26

d. 7.76 ✓

$c^2 = 12^2 + 10^2 - 2(12)(10)\cos 40^\circ \approx 244 - 183.85 \approx 60.15$, so $c \approx 7.76$.

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36 [Intermediate] • Target: Triangle congruence criteria

A triangle has sides $a = 10$, $b = 6$, and included angle $C = 70^\circ$. Find side c, to the nearest hundredth.

a. 10.74

b. 9.74 ✓

c. 9.24

d. 10.24

$c^2 = 10^2 + 6^2 - 2(10)(6)\cos 70^\circ \approx 136 - 41.04 \approx 94.96$, so $c \approx 9.74$.

37 [Intermediate] • Target: Similar triangles and proportional reasoning

A triangular plot of land has sides 24, 45, 51. Find its area using Heron's formula.

a. 542

b. 60

c. 540 ✓

d. 538

$s = \frac{24 + 45 + 51}{2} = 60$. Area $= \sqrt{60(36)(15)(9)} = \sqrt{291600} = 540$.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: The Law of Cosines • Cross-Lesson: 8.01 Statistics & Probability

A triangular plot of land has sides 36, 105, 111. Find its area using Heron's formula.

a. 1890 ✓

b. 1888

c. 126

d. 1892

$s = \frac{36 + 105 + 111}{2} = 126$. Area $= \sqrt{126(90)(21)(15)} = \sqrt{3572100} = 1890$.

39 [Challenging] • Target: Heron's formula • Cross-Lesson: 1.01 Advanced Algebra

Find the sum of the interior angles of a regular polygon with 72 sides, and the measure of each interior angle.

a. $12780, 170$

b. $12600, 175$ ✓

c. $12600, 180$

d. $12420, 175$

Sum $= (72 - 2) \times 180^\circ = 12600^\circ$. Each interior angle $= \frac{12600^\circ}{72} = 175^\circ$.

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40 [Challenging] • Target: Interior and exterior angles of polygons • Cross-Lesson: 2.01 Functions: Composite, Inverse & Logarithmic

Find the sum of the interior angles of a regular polygon with 30 sides, and the measure of each interior angle.

a. $5220, 163$

b. $5040, 168$ ✓

c. $4860, 168$

d. $5040, 173$

Sum $= (30-2) \times 180^\circ = 5040^\circ$. Each interior angle $= \frac{5040^\circ}{30} = 168^\circ$.

41 [Challenging] • Target: Coordinate Geometry of Lines • Cross-Lesson: 3.01 Sequences, Series & Complex Numbers

Find the perpendicular distance from the point $(3, -2)$ to the line $5x - 12y + 8 = 0$.

a. 2.5

b. 3.6 ✓

c. 4.0

d. 4.5

Formula: $d = \frac{|Ax_0 + By_0 + C|}{\sqrt{A^2 + B^2}} = \frac{|5(3) - 12(-2) + 8|}{\sqrt{5^2 + (-12)^2}} = \frac{|15 + 24 + 8|}{13} = \frac{47}{13} \approx 3.6$.

42 [Challenging] • Target: Coordinate Geometry of Lines • Cross-Lesson: 5.01 Trigonometry

What is the acute angle of intersection between the lines $y = 2x + 1$ and $y = -3x + 4$?

a. 45° ✓

b. 60°

c. 30°

d. 75°

Formula: $\tan\theta = |\frac{m_1 - m_2}{1 + m_1 m_2}| = |\frac{2 - (-3)}{1 + 2(-3)}| = |\frac{5}{1 - 6}| = |\frac{5}{-5}| = 1 \Rightarrow \theta = 45^\circ$.

43 [Challenging] • Target: Triangle Theorems & Inequalities • Cross-Lesson: 6.01 Circles & Coordinate Geometry

The side lengths of a triangle are $5, 12,$ and 13. What is the radius of the incircle (inradius)?

a. 1

b. 2 ✓

c. 3

d. 4

Inradius $r = \frac{A}{s}$. For a right triangle, Area $= \frac{5 \times 12}{2} = 30$. Semiperimeter $s = \frac{5 + 12 + 13}{2} = 15$. Thus $r = \frac{30}{15} = 2$.

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44 [Challenging] • Target: Polygons & Interior/Exterior Angles • Cross-Lesson: 7.01 Solid Geometry & Similarity

How many diagonals does a regular decagon (10-sided polygon) have?

a. 25

b. 35 ✓

c. 45

d. 70

Formula for diagonals: $\frac{n(n-3)}{2} = \frac{10(7)}{2} = 35$.

45 [Challenging] • Target: Geometric Transformations & Dilations • Cross-Lesson: 8.01 Statistics & Probability

A triangle with an area of 14 cm^2 is dilated by a scale factor of $k = 3$. What is the area of the image triangle?

a. 42 cm^2

b. 84 cm^2

c. 126 **cm^2** ✓

d. 168 cm^2

Area scales by k^2: $14 \times 3^2 = 14 \times 9 = 126$ cm^2.

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